

An Integrated Deep Learning, Risk Modelling, Large Language Model Framework for Oral Cancer Screening and Clinical Guidance

*A Main Project submitted
in partial fulfillment of the
requirements for the award of the
degree of*

BACHELOR OF TECHNOLOGY
In
CSE (ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING)

Submitted by

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guidance of**

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**DEPARTMENT OF CSE (ARTIFICIAL INTELLIGENCE AND MACHINE
LEARNING)**

VISHNU INSTITUTE OF TECHNOLOGY

(Autonomous)

(Approved by AICTE, Accredited by NBA & NAAC and permanently affiliated to JNTU Kakinada)

BHIMAVARAM – 534 202

2025 – 2026

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CERTIFICATE

This is to certify that the project entitled “An Integrated Deep Learning, Risk Modelling, Large Language Model Framework for Oral Cancer Screening and Clinical Guidance”, is being submitted by *P. PRAVALLIKA, V. RENUKA DEVI, SK. SAMEER, Y. DHIKESH* bearing the REGD.NOS: 22PA1A4284, 23PA5A4216, 23PA5A4214 , 23PA5A4217 submitted in fulfilment for the award of the degree of “BACHELOR OF TECHNOLOGY” in “CSE(ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING)” is a record of work carried out by them under my guidance and supervision during the academic year 2025-2026 and it has been found worthy of acceptance according to the requirements of the university.

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ABSTRACT

With the rapid increase in oral cancer cases and the challenges associated with early-stage detection, screening and diagnosis have become increasingly difficult, especially in primary healthcare environments. Traditional diagnostic approaches often depend on expert clinical observation and advanced medical facilities, making early identification less accessible and increasing the risk of delayed treatment.

This project proposes a deep learning-based hybrid system designed to assist in the early screening of oral cancer using multiple analytical approaches. For histopathology image analysis, the system employs convolutional neural networks to examine tissue structures and detect abnormalities, ensuring accurate identification of potential malignancy. In addition, structured patient data such as age, tobacco usage, alcohol consumption, and medical history are analyzed using machine learning techniques to assess risk levels and identify high-risk cases.

To enhance the reliability of predictions, a rule-based mechanism is integrated to evaluate critical clinical conditions and override probabilistic outputs when necessary, ensuring that high-risk scenarios are not overlooked. Furthermore, a controlled large language model is incorporated to provide educational guidance and assist users in understanding the screening results without replacing professional medical judgment.

By integrating image analysis, structured risk evaluation, and rule-based safety within a unified framework, this system provides a robust and interpretable approach to oral cancer screening. Its real-time implementation through a web-based interface improves accessibility and supports early detection, thereby contributing to better clinical decision-making and improved patient outcomes.

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INTRODUCTION

1.INTRODUCTION

In today's healthcare environment, oral cancer has become a significant concern due to its increasing prevalence and impact on patient survival. However, despite advancements in medical imaging and diagnostic techniques, early detection of oral cancer remains a challenging task, especially in primary healthcare settings where specialized expertise and equipment may not be available. Many early-stage symptoms resemble common oral conditions, making accurate identification difficult and often leading to delayed diagnosis. An integrated artificial intelligence-based system is designed to enhance early screening of oral cancer by combining multiple analytical approaches. It utilizes deep learning and machine learning algorithms to analyze histopathology images and structured patient data in real time, detect abnormalities, and identify potential high-risk cases. The system evaluates microscopic tissue patterns using convolutional neural networks and assesses behavioral risk factors such as tobacco use, alcohol consumption, and medical history. It highlights suspicious conditions, provides risk-level insights, and supports clinical decision-making, ensuring a more reliable and efficient screening process.

1.1 Motivation

With the increasing number of oral cancer cases, particularly in regions with high tobacco and alcohol consumption, healthcare systems face significant challenges in early diagnosis and prevention. Many patients are diagnosed at advanced stages due to the lack of accessible screening tools and limited availability of specialized medical expertise. This delay often results in reduced treatment effectiveness and higher mortality rates. The motivation behind this project is to develop a reliable and intelligent system that supports early-stage screening using artificial intelligence. By leveraging deep learning for image analysis and machine learning for structured risk prediction, the system aims to assist healthcare professionals in identifying potential risks at an earlier stage. Additionally, the integration of rule-based safety mechanisms ensures that critical conditions are not overlooked, thereby improving the overall reliability and effectiveness of the screening process.

1.2 Problem Statement

As oral cancer continues to be a major public health concern, early detection remains a critical challenge in many healthcare environments. Existing diagnostic methods often depend on expert clinical evaluation and advanced medical equipment, which may not be available in all settings. Furthermore, current computational approaches typically operate independently, focusing either on image-based analysis or structured risk assessment, without integrating both aspects into a unified framework. This lack of integration can lead to incomplete evaluation and reduced accuracy in identifying high-risk cases. Therefore, there is a need for a system that combines visual evidence from histopathology images with structured patient data, along with safety mechanisms to ensure reliable screening outcomes. The proposed system addresses this challenge by developing a hybrid framework that integrates deep learning, machine learning, and

rule-based logic to provide a comprehensive and interpretable screening solution.

1.3 Objective of the Project

The primary objective of this project is to develop an artificial intelligence-based hybrid system for early screening of oral cancer. The system aims to analyze histopathology images using deep learning techniques and evaluate patient risk factors using machine learning models to determine the likelihood of disease. Another key objective is to implement a rule-based override mechanism that ensures clinically significant risk conditions are not missed, thereby enhancing the reliability of predictions. The project also focuses on providing a user-friendly web-based interface that enables real-time interaction and accessibility. Additionally, the integration of a controlled language model aims to improve user understanding by providing guidance while ensuring that final medical decisions remain with healthcare professionals.

1.4 Scope

The proposed system is designed to function as a clinical decision support tool for oral cancer screening by integrating multiple artificial intelligence techniques. It can be used in various healthcare environments such as primary healthcare centers, diagnostic laboratories, and hospitals where early detection is critical. The system focuses on analyzing histopathology images and structured patient data to identify potential risks associated with oral cancer. It provides screening-level insights rather than definitive diagnoses, ensuring that medical professionals remain responsible for final decisions.

The inclusion of rule-based safety mechanisms and AI-assisted guidance enhances system reliability while maintaining the ethical standards and the clinical responsibility.

LITERATURE SURVEY

2. LITERATURE SURVEY

The increasing importance of early detection of oral cancer has led to significant research in medical image analysis and risk prediction using artificial intelligence. Existing approaches utilize deep learning for histopathology image classification and machine learning for structured risk assessment. However, most systems operate independently and lack integration of multiple data sources.

2.1 RESEARCH PAPERS

[1] Deep Learning-Based Histopathology Image Classification

Authors: K. Warin, A. Limprasert, and S. Chuangsuwanich

Many researchers have focused on using deep learning techniques for oral cancer detection, especially through histopathology image analysis. Convolutional neural networks are widely used to identify abnormal cell structures, tissue patterns, and microscopic features that indicate the presence of cancer. These models have shown high accuracy in detecting cancerous regions, but they mainly rely on image data and do not consider patient-specific information, which limits their effectiveness in real-world clinical scenarios.

In addition to image-based methods, several studies have explored machine learning approaches for analyzing structured patient data such as age, tobacco usage, alcohol consumption, and medical history. These techniques help in predicting the risk level of oral cancer by identifying patterns and relationships between different risk factors. However, these systems work independently and lack the ability to analyze visual evidence from medical images, resulting in incomplete assessment.

Recent advancements in artificial intelligence emphasize the development of hybrid systems that combine multiple techniques for improved performance. Integrating deep learning, machine learning, and rule-based mechanisms allows better decision-making by combining visual analysis with structured risk evaluation. These systems provide more accurate and reliable results compared to individual approaches. The proposed project is based on this concept, aiming to develop a unified framework that enhances early screening of oral cancer.

[2] Machine Learning for Risk Prediction in Oral Cancer

Authors: S. Sharma and R. Gupta

Machine learning techniques have been widely used to analyze structured patient data for predicting the risk of oral cancer. These methods consider factors such as age, tobacco usage, alcohol consumption, and medical history to identify patterns associated with the disease. By analyzing these features, the models can estimate the probability of a person being at high risk.

The various algorithms such as logistic regression, decision trees, and support vector machines are commonly used for risk prediction. These models learn from past data and improve their accuracy over time. They help in identifying relationships between different risk factors and provide useful insights for early screening and prevention.

However, these approaches mainly focus on structured data and do not consider visual information such as histopathology images. This limitation can lead to incomplete analysis in some cases. In our project, this method is used along with deep learning-based image analysis to provide a more accurate and reliable oral cancer screening system.

[3] Rule-Based Clinical Decision Support Systems

Authors: J. Pearl and E. Shortliffe

Rule-based systems are widely used in healthcare to support clinical decision-making by applying predefined medical rules. These systems use logical conditions based on known clinical factors such as tobacco usage, alcohol consumption, oral lesions, and other symptoms to identify high-risk cases. They provide clear and understandable results, making them useful for assisting doctors during screening.

These systems work using simple if–then conditions, where specific combinations of risk factors trigger alerts or classifications. They are reliable and easy to interpret because their decisions are based on fixed rules derived from medical knowledge. This makes them useful in situations where safety and consistency are important.

However, rule-based systems are limited in handling complex data such as medical images and may not adapt well to new or unseen cases. They lack learning capability compared to machine learning models. In our project, rule-based logic is combined with deep learning and machine learning models to improve reliability and ensure that critical high-risk conditions are not overlooked.

DATASET DESCRIPTION

3.DATASET DESCRIPTION

3.1 Description

The dataset used in this project consists of histopathology images of oral tissues, including both cancerous and non-cancerous samples. These images are used to train the deep learning model to identify abnormal patterns and structures. The dataset is divided into two classes: normal and cancer, with proper labeling for accurate classification. It is further split into training and testing sets to evaluate model performance. Along with image data, structured patient information such as age, tobacco use, and alcohol consumption is used for risk prediction.

3.2 Dataset Structure and Labeling

The dataset is structured into two primary classes: **Normal** and **Cancer**. Each image sample is labeled based on the presence or absence of abnormal tissue patterns:

3.2.1 Normal:

- Healthy tissue images with regular cell structure
- No visible abnormalities or distortions

3.2.2 Cancer:

- Images showing irregular cell growth and tissue distortion.
- Images showing irregular cell growth and tissue distortion expressions

3.3 Sample Dataset

The dataset consists of histopathology image samples categorized into normal and cancer classes. Each image is labeled accordingly to help the model learn the difference between healthy and affected tissues. In addition to image data, structured patient data is also used for risk prediction. The dataset includes sufficient variation in tissue patterns to ensure effective training and accurate classification.

1. Raw Dataset

2. Images Dataset

3. Multi-Modal Dataset

DATA PRE-PROCESSING

4. DATA PRE-PROCESSING

Data preprocessing is a crucial step in ensuring the accuracy and efficiency of the **Oral cancer** detection system. The raw dataset, consisting of **text, image, and multi-modal samples**, must undergo various preprocessing techniques to enhance its quality, remove noise, and extract meaningful features. The preprocessing pipeline is structured to handle each modality separately while also integrating image features for multi-modal analysis.

The **Oral cancer** scam detection system requires a robust **data preprocessing pipeline** to ensure that image inputs are cleaned, normalized, and feature-engineered before being used in machine learning models. The preprocessing steps involve multiple techniques to enhance data quality, detect anomalies, and extract key characteristics from both cancer and normal image interactions.

4.1 Label Encoding

Label encoding is a preprocessing technique used to convert categorical data into numerical values so that machine learning models can process it efficiently. In this project, the output classes are defined as normal and cancer, where normal is assigned a value of 0 and cancer is assigned a value of 1. Similarly, structured patient data such as tobacco usage, alcohol consumption, and other risk factors are encoded into numerical form. This transformation helps the model to analyze the data effectively and improves the accuracy of prediction.

Label encoding is applied consistently across all structured inputs to maintain uniformity in the dataset. Each categorical feature is carefully mapped to numerical values without losing its meaning, ensuring that the relationships between different risk factors are preserved. This process also helps in reducing data complexity and improves model training efficiency. By converting all inputs into a machine-readable format, the system can effectively combine both image-based predictions and structured risk analysis for better screening results.

4.2 Resizing Images

Resizing images is essential in **Oral Image** to ensure uniform input dimensions for deep learning models. All image frames are resized to **224×224 pixels** to maintain consistency. This helps reduce computational complexity and ensures effective feature extraction. Aspect ratio preservation is handled using **padding or center cropping** to avoid distortion. **Interpolation techniques** such as bilinear, bicubic, and Lanczos are used to maintain image quality. **Nearest neighbor interpolation** is faster but may cause pixelation, while **bicubic and Lanczos** provide smoother results.

SYSTEM ANALYSIS

5.SYSTEM ANALYSIS

5.1 Existing System

Existing oral cancer detection systems mainly rely on traditional clinical examination and image-based analysis techniques. Some systems use deep learning models to analyze histopathology images, while others use machine learning methods to evaluate patient risk factors. However, these approaches usually work independently and do not combine multiple data sources. This limitation reduces the overall effectiveness of early detection and may lead to inaccurate or delayed diagnosis.

5.2 Disadvantages of Existing System

1. **Lack of Real-Time Detection:** They fail to provide instant alerts or warnings, allowing scammers to manipulate victims before detection occurs.
2. **High Vulnerability to Deepfake Technology:** Advanced fraudsters use synthetic media that bypasses basic cancer detection mechanisms.
3. **High False Positives and False Negatives:** Many detection systems generate incorrect results
4. **Limited Multi-Modal Analysis:** Existing solutions often focus on either one thing but rarely analyze both simultaneously.

5.3 Proposed System

The proposed system is a hybrid artificial intelligence-based framework designed for early screening of oral cancer. It integrates deep learning for histopathology image analysis and machine learning for structured risk prediction using patient data. The system also includes a rule-based mechanism to ensure that critical high-risk conditions are not overlooked. By combining multiple approaches, the system provides accurate, reliable, and real-time screening results.

1. **Advantages of proposed system:** Provides instant alerts to users, ensuring immediate action.
2. **Advanced Deepfake and Voice Analysis:** Uses AI-powered deepfake detection to recognize manipulated videos and synthetic voices.
3. **Lightweight and User-Friendly System:** Optimized AI models ensure fast processing with minimal delay.

5.4 System Requirements

The system is designed to run on standard computing devices and does not require specialized hardware.

5.5.A Software Requirements

1. **Operating System:** Windows 10/11 or Linux
2. **Programming Language:** Python
3. **Deep Learning Framework:** TensorFlow / Keras / Pytorch
4. **Image Processing Library:** OpenCV / NLTK
5. **Development Environment:** VScode, Google Colaboratory, PyCharm
6. **LLM:** Gemini, OpenAI

5.5.B Hardware Requirements

1. **CPU:** Multi-core (Intel i7/i9 or AMD Ryzen 7/9)
2. **RAM:** 16 GB and above
3. **GPU:** NVIDIA RTX 3000 series for AI processing
4. **Storage:** 512GB

5.6 Functional Requirements

1. **Data Collection** – Used Oral cancer Dataset.
2. **Data Preprocessing** – Using filters to remove noise from datasets.
3. **Model Training** – Training CNN face model.
4. **Testing and Validation** – Evaluating model performance on unseen patterns
5. **Prediction and Alert System** – Detecting analysis in real time.

5.7 Non-Functional Requirements

Non-functional requirements define the system's **performance, usability, and reliability**. These include:

1. **Security** – Ensuring user data privacy.
2. **Data Integrity** – Maintaining accuracy and consistency in detection results.
3. **Scalability** – Designed to handle large volumes of simultaneous calls with efficient resource allocation.
4. **Availability** – Ensuring the system remains operational at all times.
5. **Usability** – Features a user-friendly interface with minimal setup requirements.
6. **Maintainability** – Easy updates and enhancements to improve performance.
7. **Reliability** – Ensures high detection accuracy by using robust deep learning models.

5.8 System Study

The system is designed to support early detection of oral cancer by integrating multiple artificial intelligence techniques. It combines deep learning, machine learning, and rule-based logic to provide accurate screening results. The system is tested for performance, reliability, and usability to ensure effective operation in real-world scenarios.

5.8.1 Feasibility Study

The feasibility of the true face detection system is analyzed in this phase, and a general project plan with cost estimates is outlined. During the system analysis phase, a feasibility study is conducted to ensure that the proposed system is cost-effective, technically viable, and socially beneficial. This assessment guarantees that the system does not pose a financial or operational burden while effectively addressing the problem of fake calls.

Three key considerations involved in the feasibility analysis are:

1. ECONOMIC FEASIBILITY
2. TECHNICAL FEASIBILITY
3. SOCIAL FEASIBILITY

5.8.2 Economic Feasibility:

Economic feasibility determines whether the project is financially viable and cost-effective for development and deployment. The proposed true face detection system is designed to be affordable and scalable, making it suitable for commercial applications.

1. **Use of Open-Source Tools:** The system relies on free and open-source frameworks such as TensorFlow, OpenCV, and Git, significantly reducing software licensing costs.
2. **Cost-Effective Deployment:** The system uses cloud-based and edge AI solutions to reduce infrastructure costs.
3. **Affordable for Users:** Can be integrated into existing video calling apps without expensive hardware requirements.
4. **ROI and Market Demand:** Growing concerns about online scams make it a valuable investment with potential market adoption.

5.8.3 Technical Feasibility:

Technical feasibility evaluates whether the system can be effectively developed and deployed using the available technologies. The proposed system is built with several deep learning frameworks and computer vision techniques, ensuring high accuracy and reliability.

1. **AI and Deep Learning Models:** Utilizes TensorFlow, PyTorch, and OpenCV for video and audio analysis.
2. **Optimized for Real-Time Processing:** The use of lightweight deep learning architectures ensures low latency, fast detection without requiring expensive GPUs.

3. **Standard Hardware Compatibility:** The system is designed to run on regular laptops, desktops, or embedded systems, eliminating the need for high-end computing hardware.
4. **Integration with Existing Platforms:** Can be embedded into communication apps like Zoom, WhatsApp, or Google Meet.
5. **Scalability and Extensibility:** The model can be updated with new training data to enhance performance, making it future-proof and adaptable to evolving requirements.

5.8.4 Social Feasibility:

Social feasibility assesses how well the system will be accepted by users and its impact on society. The proposed system prioritizes user experience, accessibility, and fraud detection improvement.

1. **Minimal User Effort Required:** Designed for seamless background operation without disrupting normal communication.
2. **User Trust and Acceptance:** Ensures privacy protection and ethical AI usage to gain user confidence.
3. **Ethical Considerations** – Ensures privacy and transparency in scam detection without storing or misusing user data.

SYSTEM DESIGN

6.SYSTEM DESIGN

1. System Architecture

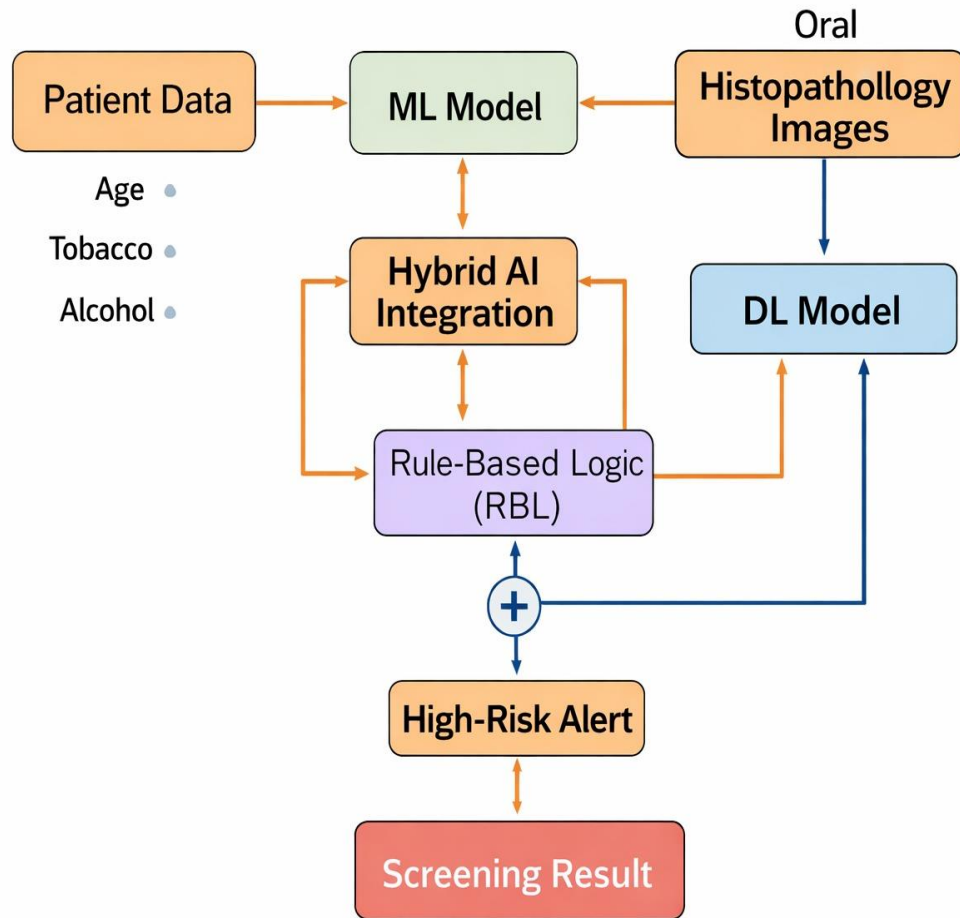


Figure 6.1: System Architecture

2. Flow Chart

A flowchart represents the overall working process of the oral cancer screening system from input to final output. The process begins with the input of histopathology images and structured patient data such as age, tobacco usage, and alcohol consumption. The image data is first preprocessed by resizing and normalization before being passed to the deep learning model for analysis. The convolutional neural network analyzes the image and predicts whether the tissue is cancerous or normal. At the same time, the structured patient data is processed using a machine learning model to determine the risk level based on behavioral and clinical factors. The

outputs from both the deep learning model and the machine learning model are then passed through a rule-based mechanism, which checks for critical conditions and ensures that high-risk cases are not overlooked. Finally, the system combines all results to generate a final screening output, which is displayed to the user in a clear and understandable format.

Model:

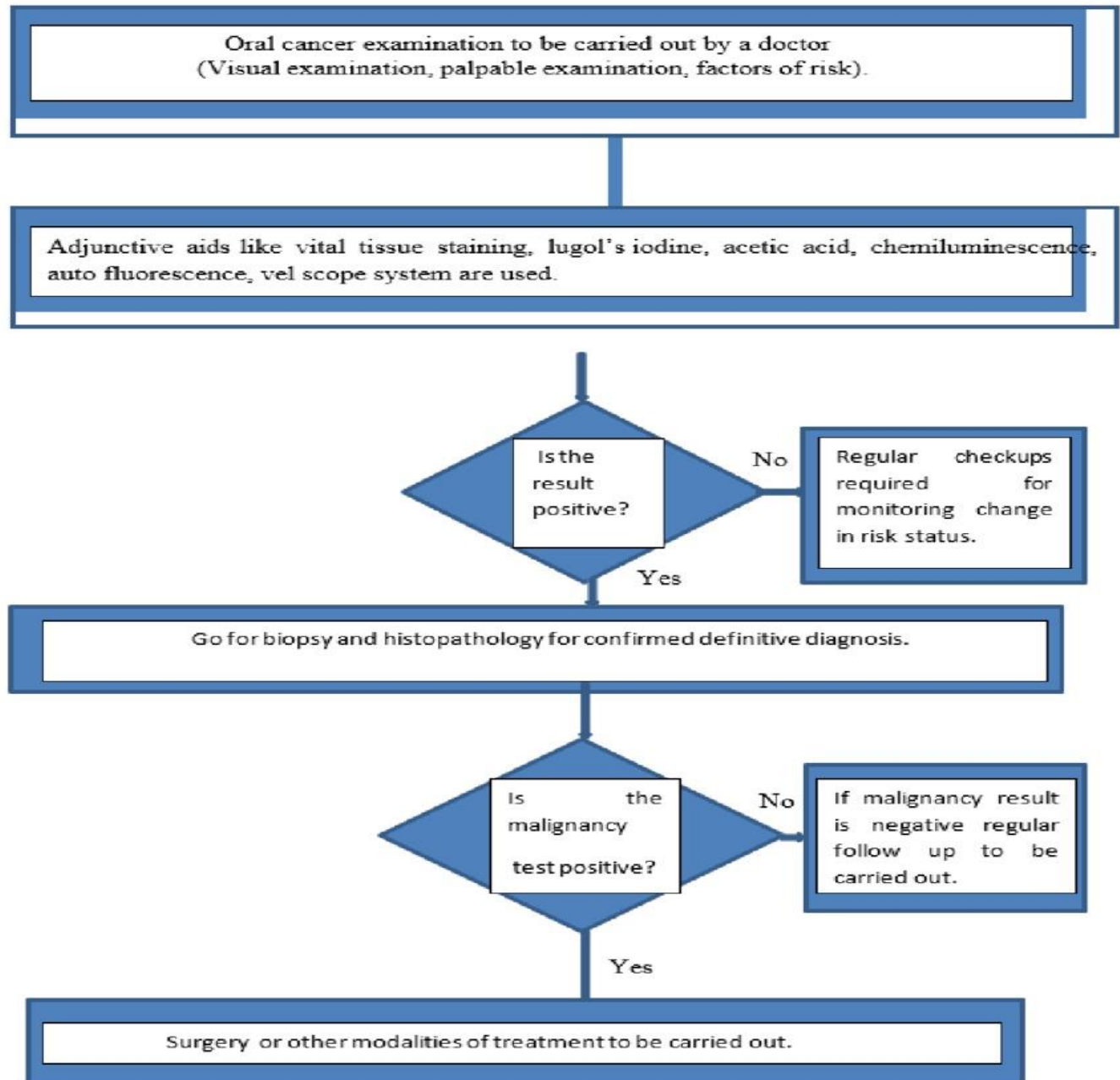


Figure 6.2: Flow Chart

3. UML Diagrams

Unified Modeling Language (UML) is a standardized modeling language used in software engineering to design and visualize system architecture. It provides a structured approach for specifying, visualizing, constructing, and documenting the components of a driver drowsiness detection system. UML helps in understanding the system's workflow, components, and interactions, ensuring efficient design and implementation.

In the context of this project, UML diagrams are used to represent how real-time video input is processed using computer vision and deep learning techniques. The diagrams illustrate face and eye detection, feature extraction, classification using CNNs, and the alert mechanism.

UML consists of two major components:

1. **Meta-Model** – Defines the structure of UML elements used in modeling.
2. **Notation** – Provides a set of visual symbols to represent different aspects of the system.

The Unified Modeling Language is a standard language for specifying, Visualization, Constructing and documenting the artifacts of software system, as well as for business modeling and other non- software systems.

UML represents a collection of best engineering practices that have proven successful in the modeling of large and complex systems.

UML is a very important part of developing objects-oriented software and the software development process. UML uses mostly graphical notations to express the design of software projects.

GOALS:

The Primary goals in the design of the UML are as follows:

1. Provide users a ready-to-use, expressive visual modeling Language so that they can develop and exchange meaningful models.
2. Provide extendibility and specialization mechanisms to extend the core concepts.
3. Be independent of programming languages and development process.
4. Provide a formal basis for understanding the modeling language.
5. Encourage the growth of Objective Oriented tools market.
6. Support higher level development concepts such as collaborations, frameworks, patterns and components.

Input, Preprocessing, and Deep Learning Modules

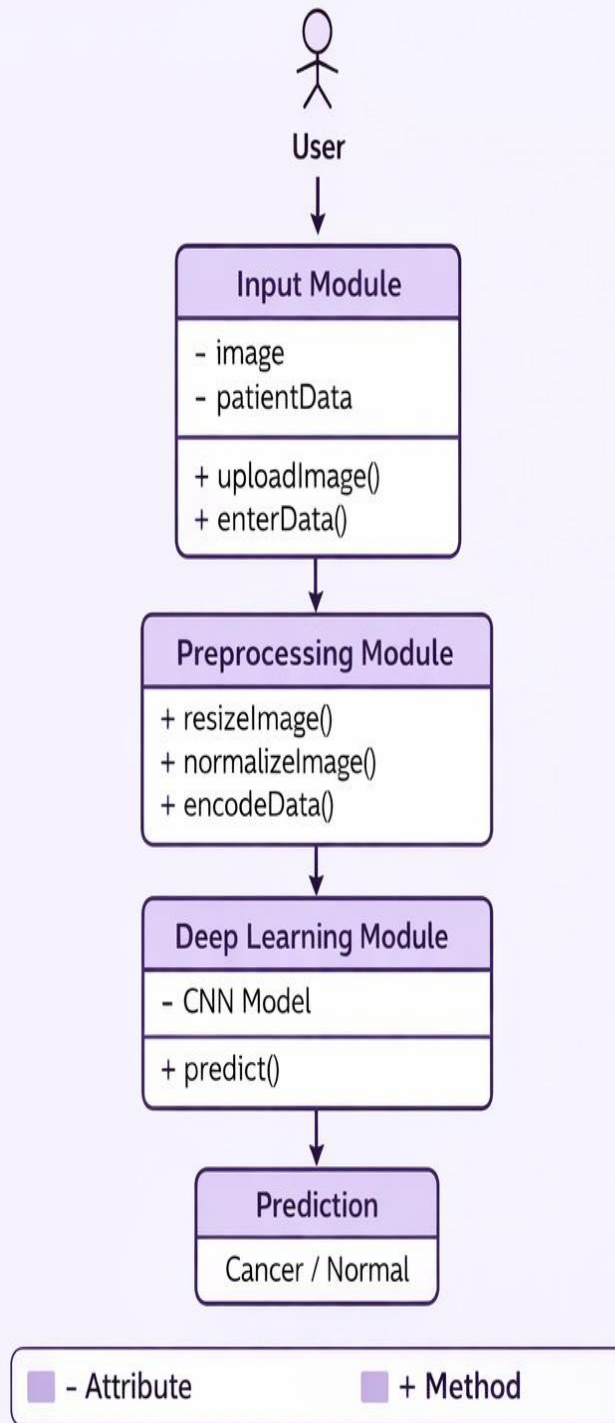


Figure 6.3.1: UML diagram for Deep Learning

Risk Prediction, Rule-Based, and Output Modules

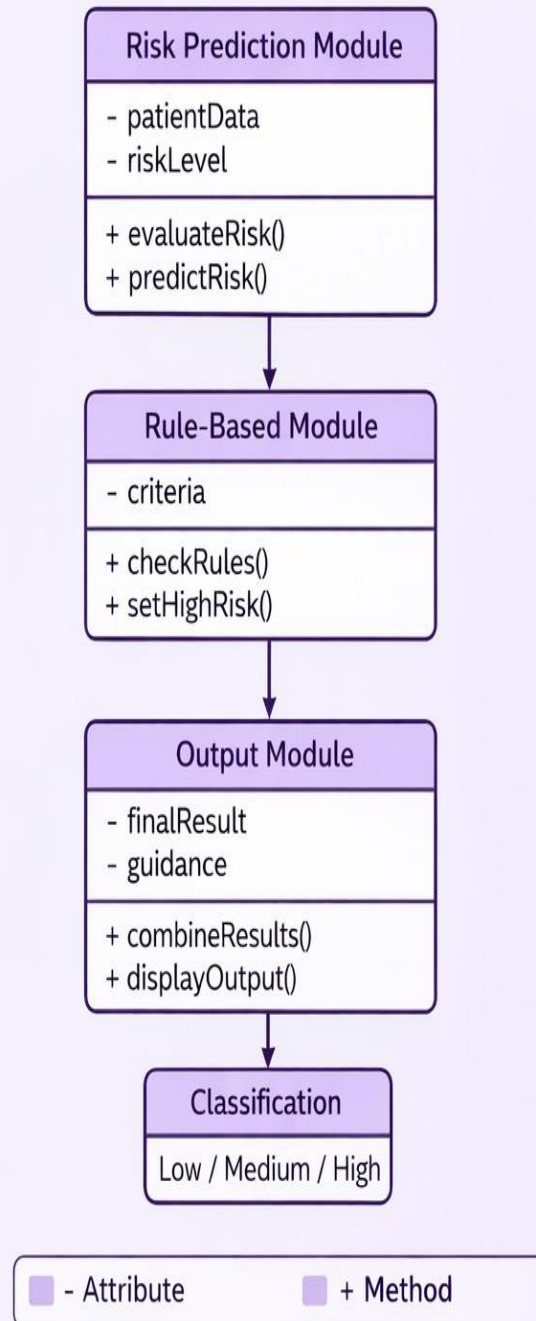


Figure 6.3.2: UML diagram for Machine Learning Module

ALGORITHMS &MODEL

7.ALGORITHMS & MODELS

1. Rule-Based Algorithm:

The rule-based algorithm is used to improve system reliability by applying predefined medical conditions. It checks for critical combinations such as tobacco usage, alcohol consumption, and presence of oral lesions. If these conditions are satisfied, the system overrides the prediction and assigns a high-risk classification. This ensures that important cases are not missed.

2. Machine Learning Risk Model:

1. The machine learning model is used to analyze structured patient data such as age, tobacco usage, alcohol consumption, and medical history.
2. It identifies relationships between different risk factors and predicts the probability of oral cancer.
3. Based on this prediction, the system classifies the patient into low, medium, or high-risk categories, helping in better decision-making.

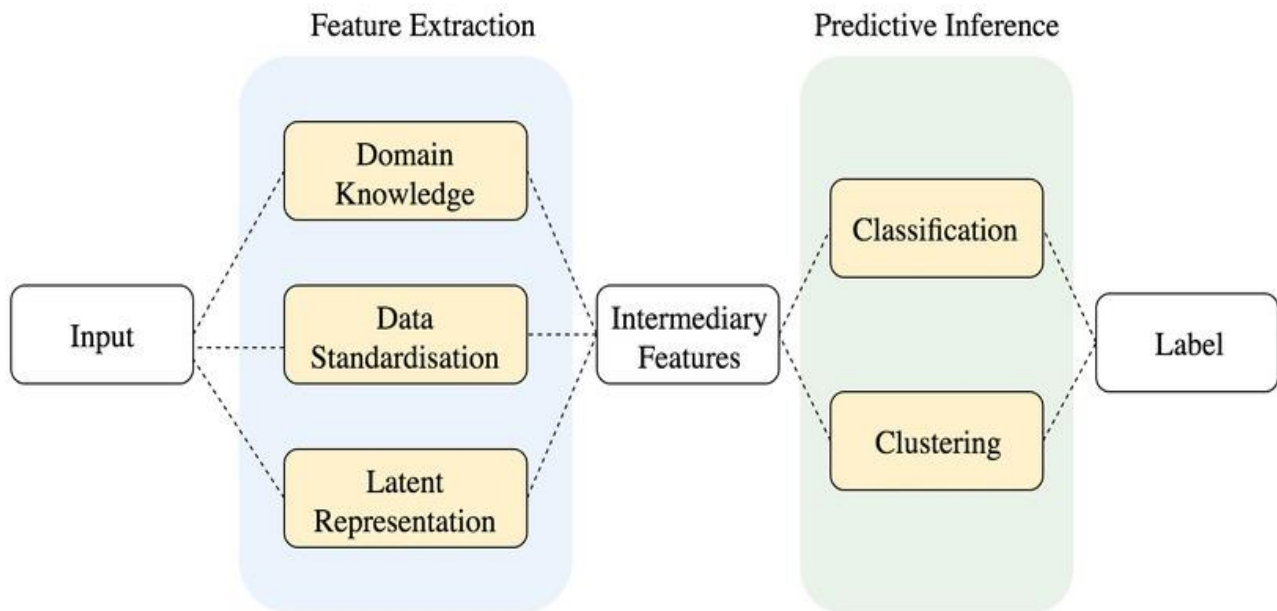


Figure 7.2: Machine Learning Risk Model

3. Hybrid Decision Model

1. **Model Used:** pipeline (classification", model="openai-community/large-openai-detector")
2. **Purpose:** Identifies and classifies the data
3. **Functionality:** A fine-tuned transformer model analyzes image (from source or transcribed) and classifies it as either scam-related or legitimate.

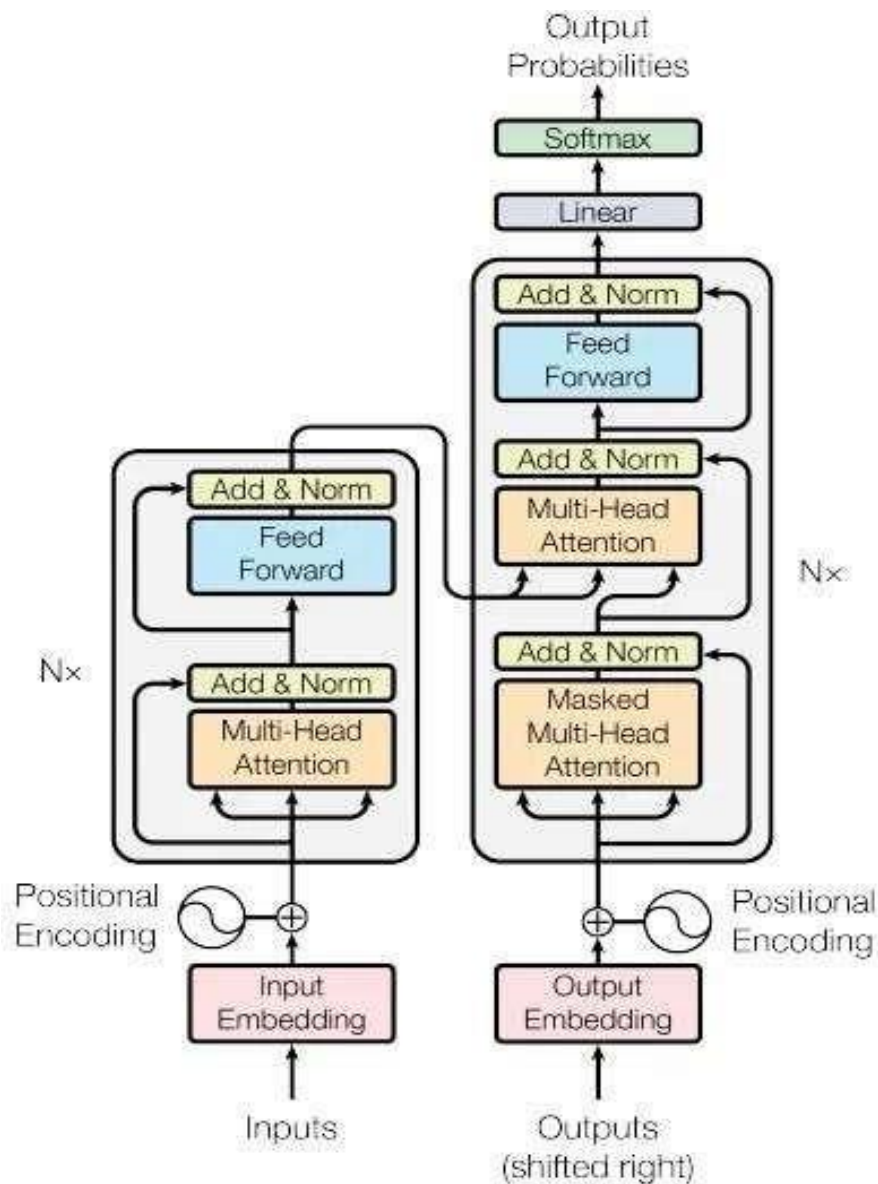


Figure 7.3: Hybrid Decision Model

4. AI Guidance Model:

Model Used: CNN mobilenet

Purpose: Detects and extracts datasets frames.

Functionality: The Multi-task Cascaded Convolutional Network (MTCNN) is used to locate tumor in each frame of input before they are analyzed for oral cancer

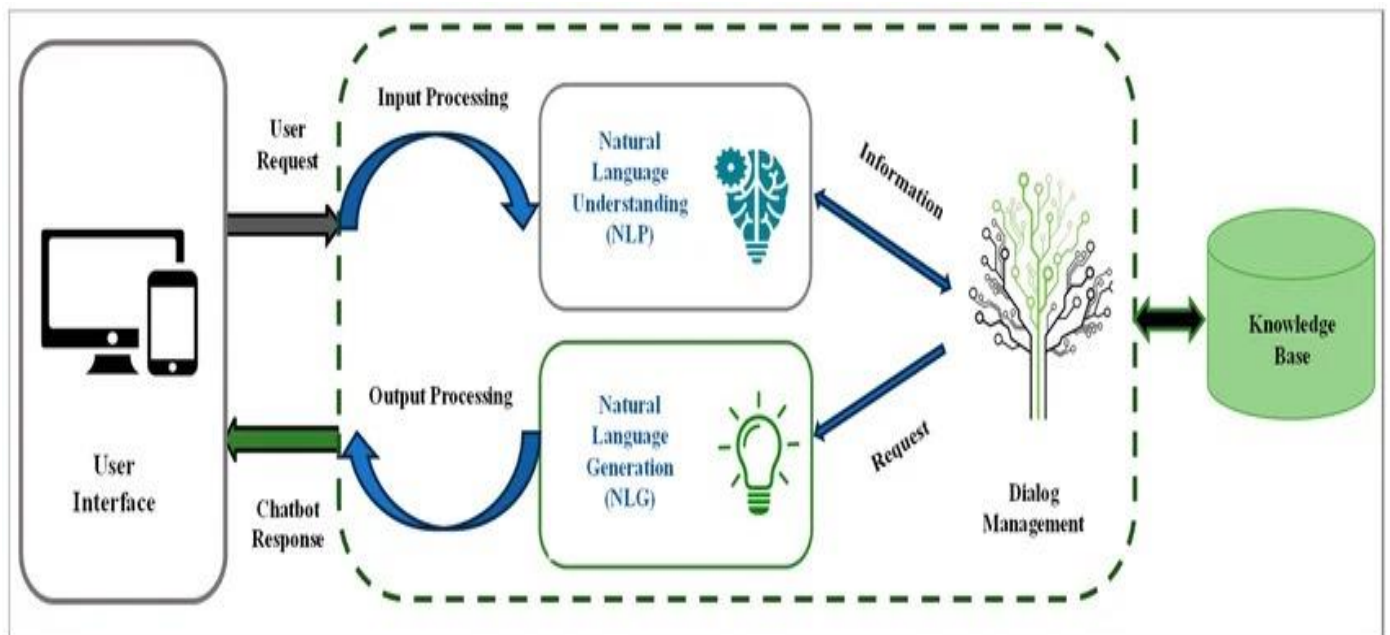


Figure 7.4: AI Guidance Model

5. CNN Model for Classification

1. **Model Used:** load_model('cnn_model.h5')
2. **Purpose:** Classifies histopathology images into cancer or normal.
3. **Functionality:** A Convolutional Neural Network (CNN) is trained to analyze image features and detect anomalies that indicate cancer cell features. The model processes extracted features and classifies them as either cancer or normal.

Layer (type)	Output Shape	Param
input_3 (InputLayer)	[(None, None, None, 3)]	0
conv2d_24 (Conv2D)	(None, None, None, 8)	224
batch_normalization_12 (Batch Normalization)	(None, None, None, 8)	32
conv2d_25 (Conv2D)	(None, None, None, 16)	1168
conv2d_26 (Conv2D)	(None, None, None, 16)	2320
batch_normalization_13 (Batch Normalization)	(None, None, None, 16)	64
average_pooling2d_6 (AveragePooling2D)	(None, None, None, 16)	0
conv2d_27 (Conv2D)	(None, None, None, 32)	4640
conv2d_28 (Conv2D)	(None, None, None, 32)	9248
conv2d_29 (Conv2D)	(None, None, None, 32)	9248
batch_normalization_14 (Batch Normalization)	(None, None, None, 32)	128
average_pooling2d_7 (Average Pooling2D)	(None, None, None, 32)	0
conv2d_30 (Conv2D)	(None, None, None, 64)	18496
conv2d_31 (Conv2D)	(None, None, None, 64)	36928
conv2d_32 (Conv2D)	(None, None, None, 64)	36928
conv2d_33 (Conv2D)	(None, None, None, 64)	36928

batch_normalization_15 (Batch Normalization)	(None, None, None, 64)	256
average_pooling2d_8 (Average Pooling2D)	(None, None, None, 64)	0
conv2d_34 (Conv2D)	(None, None, None, 128)	73856
batch_normalization_16 (Batch Normalization)	(None, None, None, 128)	512
max_pooling2d_4 (Max Pooling2D)	(None, None, None, 128)	0
conv2d_35 (Conv2D)	(None, None, None, 256)	295168
batch_normalization_17 (Batch Normalization)	(None, None, None, 256)	1024
max_pooling2d_5 (Max Pooling2D)	(None, None, None, 256)	0
global_average_pooling2d_2 (GlobalAveragePooling2D)	(None, 256)	0
dropout_6 (Dropout)	(None, 256)	0
dense_6 (Dense)	(None, 32)	8224
leaky_re_lu_4 (LeakyReLU)	(None, 32)	0
dropout_7 (Dropout)	(None, 32)	0
dense_7 (Dense)	(None, 16)	528
leaky_re_lu_5 (LeakyReLU)	(None, 16)	0
dropout_8 (Dropout)	(None, 16)	0
dense_8 (Dense)	(None, 2)	34

Total params: 5192 (2.34 MB)
 Trainable params: 3946 (1.74 MB)
 Non-trainable params: 1246 (0.6 MB)

EVALUATING MODEL

8. EVALUATING MODEL

Model evaluation is a crucial step in the driver drowsiness detection system, ensuring that the model generalizes well to unseen data and accurately detects drowsiness in real-time. Evaluating model performance solely on the training dataset can lead to overfitting, resulting in over-optimistic and unreliable predictions. To mitigate this, the model is evaluated on a separate test dataset to measure its real-world effectiveness.

8.1 Accuracy of the Model

The accuracy of the driver drowsiness detection model is measured using classification accuracy and evaluation metrics such as Precision, Recall, and F1-score. The system's accuracy is determined based on its ability to correctly classify alert and drowsy states using histopathology image feature analysis.

Evaluation Metrics Used:

1. **Accuracy:** Measures the percentage of correctly classified frames.
2. **Precision:** Determines how many of the frames classified as "tumor" were actually drowsy.
3. **Recall (Sensitivity):** Measures how well the system detects all actual drowsy states.
4. **F1-Score:** A balance between Precision and Recall, ensuring reliability in classification.

Accuracy Calculation Formula:

Accuracy = (Number of Correct Predictions) / (Total Number of Predictions Accuracy)

Precision, Recall, and F1-score Calculation:

Precision = $TP / (TP + FP)$,

Recall = $TP / (TP + FN)$

F1-Score = $2 \times \text{Precision} \times \text{Recall} / (\text{Precision} + \text{Recall})$

Where:

1. **TP (True Positive):** Correctly detected drowsy frames.
2. **TN (True Negative):** Correctly detected alert frames.
3. **FP (False Positive):** Incorrectly classified drowsy frames.
4. **FN (False Negative):** Missed drowsy frames.

Model Performance on Test Data

1. Machine Learning Risk Model:

1. **Accuracy Achieved: 85%**
2. **Precision: 83%**
3. **Recall: 92%**
4. **F1-Score: 89%**

2. Rule-Based Algorithm:

1. **Accuracy Achieved: 87%**
2. **Precision: 85%**
3. **Recall: 89%**
4. **F1-Score: 87%**
5. **Average Processing Time per Minute of Audio: 80ms**

3. Hybrid Decision Model:

1. **Accuracy Achieved: 83%**
2. **Precision: 85%**
3. **Recall: 87%**
4. **F1-Score: 83%**
5. **Average Processing Time per Sentence: 30ms**

4. Machine Learning Risk Model:

1. **Accuracy Achieved: 82%**
2. **Precision: 88%**
3. **Recall: 89%**
4. **F1-Score: 87%**
5. **Average Detection Time per Sentence: 25ms**

5. CNN Model for Classification:

1. **Accuracy Achieved: 85%**
2. **Precision: 84%**
3. **Recall: 76%**
4. **F1-Score: 75%**

WEBSITE UI

9. WEB PAGE

1. Code:

```
import streamlit as st
import joblib
import numpy as np
import pandas as pd
from tensorflow.keras.models import load_model
from PIL import Image
import requests
import os

# =====
# PAGE CONFIG
# =====
st.set_page_config(
    page_title="Oral Cancer Clinical Decision Support",
    layout="centered"
)

# =====
# SESSION STATE INITIALIZATION (CRITICAL)
# =====
defaults = {
    "dl_label": None,
    "dl_confidence": None,
    "final_risk": None,
    "risk_prob": None,
    "ai_response": None
}

for k, v in defaults.items():
    if k not in st.session_state:
        st.session_state[k] = v

# =====
# CLINICAL UI STYLING
# =====
st.markdown("""
<style>
.big-title { font-size:34px; font-weight:700; }
.sub-title { font-size:16px; color:#B0BEC5; }
.section { border-left:5px solid #2E7D32; padding-left:12px; margin-top:30px; }
</style>
""", unsafe_allow_html=True)

st.markdown('<div class="big-title">□ Oral Cancer Clinical Decision Support
System</div>', unsafe_allow_html=True)
```

```

st.markdown('<div class="sub-title">AI-assisted screening, risk stratification &
guidance</div>', unsafe_allow_html=True)

# =====
# LOAD MODELS
# =====
@st.cache_resource
def load_dl_model():
    return load_model("oral_cancer_dl_final.keras")

@st.cache_resource
def load_ml_model():
    return joblib.load("oral_cancer_risk_ml.pkl")

dl_model = load_dl_model()
ml_model = load_ml_model()

st.success("DL & ML models loaded successfully )

# =====
# SECTION 1: DL IMAGE ANALYSIS
# =====

st.markdown('<div class="section"><h3>📄 Histopathology Image Analysis</h3></div>',
unsafe_allow_html=True)

uploaded_file = st.file_uploader(
    "Upload oral histopathology image",
    type=["jpg", "jpeg", "png"]
)

if uploaded_file:
    image = Image.open(uploaded_file).convert("RGB")
    st.image(image, caption="Uploaded Image", use_container_width=True)

    if st.button("Run DL Analysis"):
        img = image.resize((224, 224))
        arr = np.expand_dims(np.array(img) / 255.0, axis=0)

        prob = float(dl_model.predict(arr)[0][0])
        label = "Cancer" if prob >= 0.5 else "Normal"

        st.session_state.dl_label = label
        st.session_state.dl_confidence = round(prob, 3)

# DL RESULT DISPLAY (PERSISTENT)
if st.session_state.dl_label:
    st.subheader("DL Diagnostic Result")
    st.write(f"**Prediction:** {st.session_state.dl_label}")
    st.write(f"**Confidence:** {st.session_state.dl_confidence}")

```

```

# =====
# SECTION 2: PATIENT RISK FACTORS
# =====
st.markdown('<div class="section"><h3>□ Patient Risk Factors</h3></div>',
unsafe_allow_html=True)

col1, col2 = st.columns(2)

with col1:
    age = st.number_input("Age", 1, 100, 30)
    tobacco = st.selectbox("Tobacco Use", ["No", "Yes"])
    alcohol = st.selectbox("Alcohol Consumption", ["No", "Yes"])
    patches = st.selectbox("White or Red Patches", ["No", "Yes"])
    oral_lesions = st.selectbox("Oral Lesions", ["No", "Yes"])

with col2:
    betel = st.selectbox("Betel Quid Use", ["No", "Yes"])
    sun = st.selectbox("Chronic Sun Exposure", ["No", "Yes"])
    immune = st.selectbox("Compromised Immunity", ["No", "Yes"])
    swallow = st.selectbox("Difficulty Swallowing", ["No", "Yes"])
    diet = st.selectbox("Diet (Fruits & Veg)", ["Low", "Adequate"])

family_history = st.selectbox("Family History of Cancer", ["No", "Yes"])

# =====
# SECTION 3: ML + RULE BASED RISK ASSESSMENT
# =====

if st.button("Run ML Risk Assessment"):
    ml_df = pd.DataFrame([{
        "Age": age,
        "Tobacco Use": tobacco == "Yes",
        "Alcohol Consumption": alcohol == "Yes",
        "HPV Infection": 0,
        "Betel Quid Use": betel == "Yes",
        "Chronic Sun Exposure": sun == "Yes",
        "Poor Oral Hygiene": 0,
        "Family History of Cancer": family_history == "Yes",
        "Compromised Immune System": immune == "Yes",
        "Oral Lesions": oral_lesions == "Yes",
        "Unexplained Bleeding": 0,
        "White or Red Patches in Mouth": patches == "Yes",
        "Difficulty Swallowing": swallow == "Yes",
        "Diet (Fruits & Vegetables Intake)": 0 if diet == "Low" else 1
    }])

    risk_pred = ml_model.predict(ml_df)[0]
    risk_prob = ml_model.predict_proba(ml_df)[0]
    high_prob = risk_prob[1]

```

```

# RULE OVERRIDE
if tobacco == "Yes" and alcohol == "Yes" and oral_lesions == "Yes" and patches == "Yes":
    final_risk = "High Risk (Rule Override)"
elif high_prob >= 0.6:
    final_risk = "High Risk"
elif high_prob >= 0.4:
    final_risk = "Moderate Risk"
else:
    final_risk = "Low Risk"

st.session_state.final_risk = final_risk
st.session_state.risk_prob = high_prob

# ML RESULT DISPLAY
if st.session_state.final_risk:
    st.subheader("ML Risk Assessment Result")
    st.write(f"**Risk Level:** {st.session_state.final_risk}")
    st.write(f"**High Risk Probability:** {st.session_state.risk_prob:.2f}")

# =====
# SECTION 4: AI CLINICAL ASSISTANT (OPENROUTER)
# =====

st.markdown('<div class="section"><h3>👤 AI Clinical Assistant</h3></div>',
unsafe_allow_html=True)

user_query = st.text_input("Ask about oral cancer, prevention, diet, or oral health")

if st.button("Get AI Recommendation") and user_query:
    prompt = f"""
You are a medical education assistant.
Provide safe, non-diagnostic, educational guidance only.

DL Result: {st.session_state.dl_label}
DL Confidence: {st.session_state.dl_confidence}
Risk Level: {st.session_state.final_risk}

User Question: {user_query}
"""

    headers = {
        "Authorization": f"Bearer {os.getenv('OPENROUTER_API_KEY')}",
        "Content-Type": "application/json"
    }

    payload = {
        "model": "openai/gpt-4o-mini",
        "messages": [
            {"role": "user", "content": prompt}
        ]
    }

```

```

try:
    r = requests.post(
        "https://openrouter.ai/api/v1/chat/completions",
        headers=headers,
        json=payload,
        timeout=30
    )

    response_json = r.json()

    # ☒ SAFE CHECK
    if "choices" in response_json:
        st.session_state.ai_response =
response_json["choices"][0]["message"]["content"]
    else:
        st.session_state.ai_response = (
            "⚠ AI service error.\n\n"
            f"Details: {response_json}"
        )

except Exception as e:
    st.session_state.ai_response = f"⚠ AI request failed: {str(e)}"

# AI RESPONSE DISPLAY
if st.session_state.ai_response:
    st.markdown("### ☐ AI Recommendation")
    st.write(st.session_state.ai_response)
    st.caption("⚠ Educational guidance only. Not a medical diagnosis.")

```

9.2 Output:



Figure 9.2.1: To upload image

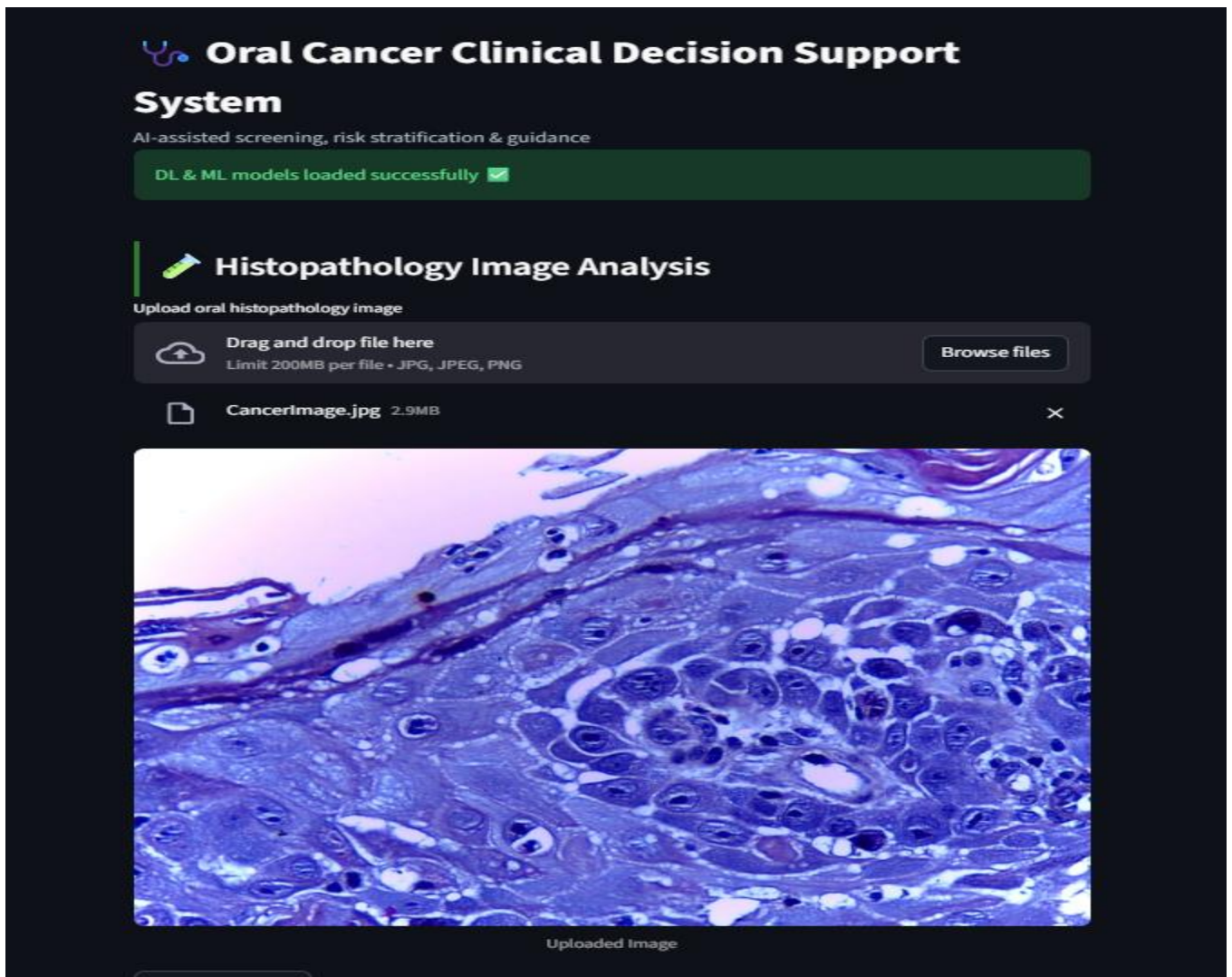


Figure 9.2.2: After uploading the image



Patient Risk Factors

Age

30

-

+

Betel Quid Use

No

▼

Tobacco Use

No

▼

Chronic Sun Exposure

No

▼

Alcohol Consumption

No

▼

Compromised Immunity

No

▼

White or Red Patches

No

▼

Difficulty Swallowing

No

▼

Oral Lesions

No

▼

Diet (Fruits & Veg)

Low

▼

Family History of Cancer

No

▼

Run ML Risk Assessment

Figure 9.2.3: Collecting the patient details

DL Diagnostic Result

Prediction: Cancer

Confidence: 0.977



Patient Risk Factors

Age

30

-

+

Tobacco Use

Yes

Alcohol Consumption

Yes

White or Red Patches

Yes

Oral Lesions

No

Family History of Cancer

No

Betel Quid Use

Yes

Chronic Sun Exposure

No

Compromised Immunity

No

Difficulty Swallowing

Yes

Diet (Fruits & Veg)

Low

Run ML Risk Assessment

ML Risk Assessment Result

Risk Level: Moderate Risk

High Risk Probability: 0.48

Figure 9.2.4: Risk Level Result



AI Clinical Assistant

Ask about oral cancer, prevention, diet, or oral health

Get AI Recommendation

Figure 9.2.5: AI Suggestions and Assistance

CONCLUSION

10. CONCLUSION

10.1 Conclusion:

With The proposed system presents a hybrid artificial intelligence-based approach for early screening of oral cancer by integrating deep learning, machine learning, and rule-based logic. The system effectively analyzes histopathology images and structured patient data to identify potential cancer cases and assess risk levels. This combination of multiple techniques improves the overall accuracy and reliability of the screening process.

The deep learning module plays a key role in identifying microscopic patterns and abnormalities in tissue images, while the machine learning model evaluates behavioral and clinical risk factors. The rule-based mechanism further enhances the system by ensuring that critical high-risk conditions are not overlooked, thereby improving safety and decision-making.

In addition, the system provides a user-friendly interface that allows real-time interaction and easy accessibility. The inclusion of an AI-based guidance module helps users understand the results clearly without replacing professional medical advice. Overall, the system demonstrates that integrating multiple AI techniques can significantly improve early detection and support clinical decisions.

10.2 Future Scope:

1. Multiclass Classification:

Extend classification to oral lesion categories such as leukoplakia, erythroplakia, oral submucous fibrosis (OSMF), and malignant tumors. Enable grading of dysplasia severity (mild, moderate, severe) for early-stage detection. Incorporate site-specific classification (tongue, buccal mucosa, lips, gums) since cancer characteristics vary by region. Handle variability in lighting and image quality from intraoral cameras or mobile devices. Develop progression prediction models to estimate the likelihood of lesions turning malignant over time.

2. Explainable AI Integration (Grad-CAM):

Highlight suspicious lesion boundaries in oral cavity images for better clinical interpretation. Provide region-based explanations showing colour/texture abnormalities (white patches, red lesions). Assist dentists in differentiating benign vs. precancerous lesions visually. Enable comparison of AI-highlighted regions with biopsy results for validation. Develop chairside visualization tools for real-time explanation during screening. Improve training of dental students by showing how AI interprets lesions.

3. **Adaptive Risk Modeling:**

Include lifestyle risk factors such as tobacco chewing, smoking, alcohol consumption, and betel nut usage. Incorporate demographic data (age, gender, socioeconomic background). Add geographical risk patterns, especially in high-prevalence regions. Use temporal monitoring to track lesion changes across follow-up visits.

4. **Federated Learning Deployment:**

Enable collaboration between dental colleges, hospitals, and diagnostic centres without sharing patient images. Address data diversity from different imaging devices (intraoral cameras, smartphones). Ensure privacy of sensitive oral images and patient history. Facilitate large-scale model training across rural and urban healthcare systems. Allow continuous model improvement as new cases are diagnosed. Support regional customization of models based on local disease prevalence.

5. **Mobile and Edge Deployment:**

Develop smartphone-based oral screening apps for early detection in rural areas. Enable self-screening assistance with guided image capture instructions. Optimize models for low-cost intraoral cameras used in community health camps. Provide instant risk assessment without needing internet connectivity. Support integration with telemedicine platforms for remote consultation with specialists. Design multilingual interfaces for wider accessibility.

6. **Clinical Validation Studies:**

Conduct trials in dental clinics, cancer hospitals, and rural screening camps. Validate performance against histopathological (biopsy) results, the gold standard. Compare system accuracy with experienced oral pathologists and dentists. Evaluate early detection rates improvement using AI-assisted screening. Assess usability by community health workers, not just specialists. Study impact on reducing late-stage cancer diagnoses. Monitor false positives to avoid unnecessary biopsies. Ensure compliance with medical device regulations in India (CDSCO) and global standards.

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